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Deep Learning Approaches for Road Damage Detection Using YOLOv8 and Custom Convolutional Neural Network

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***Abstract*—The identification of road damage is deemed very important to the preservation of infrastructure. Since recent development in deep learning provide great potential approaches, in this paper, three deep learning models, namely EfficientNet, CNN and YOLOv8 are considered and their performance is evaluated on a dataset with 8,586 images with labelled road damages consisting of seven subclasses including longitudinal cracks and potholes. Besides, other issues like class imbalance and variabilities in the environment were studied and considered in its assessment based on the Models accuracy, precision, recall, and F1-score. With an accuracy of 64%, YOLOv8 is the most accurate model for identifying various types of damages, particularly in identifying multiple damages present in one or more types of damage, as was seen in the high accuracy rate of a split sample of 44 images of multiple damages. Our findings showcase YOLOv8 as a strong contender for real-time road damage detection with potential of enhancement with better augmentation and fine-tuning.**

***Index Terms*—Deep Learning, Road Damage Detection, YOLOV8, EfficientNet, Convolutional Neural Networks (CNN), Computer Vision**

I. INTRODUCTION

Pothole intervention is a major problem experienced by transport bureaus globally, reducing vehicle stability, and enhancing expenses on repair and maintenance. Various strategies for inspecting roads usually are long, exhausting and require much energy, which explains the necessity to employ automatic ways of identifying road damage. With the optimization of the computer vision approaches and deep learning algorithms it is feasible to automate the detection and classification of road damage types.

Automated road damage detection using deep learning has made significant progress in recent years, with various models showing great promise in identifying different types of damage. But even with these advancements, challenges still persist. Many existing models struggle when it comes to accurately detecting smaller or more intricate types of damage, especially when the environment changes. Another issue is class imbalance, where certain types of damage, like small cracks, are much less common in the data. This imbalance often causes models to perform better on more frequent issues, like potholes, while overlooking the less frequent ones.

In this paper, we tackle these issues by comparing three modern deep learning models: EfficientNet, a custom-built Convolutional Neural Network (CNN), and YOLOv8. What makes our work unique is that we focus on detecting smaller and more complex road damages while also addressing the problems caused by class imbalance and environmental variations. Our findings show that YOLOv8 outperforms the other models in terms of accuracy and robustness, even in challenging conditions like low contrast or noisy settings.

II. Related Work

UAVs used for data collection and deep learning are one of the first approaches to automated road damage detection. Arya et al. [1] proposed an idea where UAVs are used to take photographs of the roads and use a deep learning model to recognize damage. The performance of this system encouraged large scale activities typically involving tedious manual inspections.

Recently, Diao et al. [2] have introduced a **Light-weight Efficient YOLO v5**, a YOLOv5 version that is tailored for use in mobile devices and edge computing infrastructure. In LE-YOLOv5, thus, special attention mechanisms were implemented to enhance the detection results, specifically for the small cracks and other less significant damages.

To enhance the detection accuracy specifically in the environments of low contrast, **TransCrack** was introduced by Lin et al. [3]. The fine-grained cracks and other types of road damages were also accurately identified with the help of self-attention mechanisms implemented in this transformer-based model. TransCrack was able to incorporate both local and global features to achieve a better detection rate for potential damages than traditional CNNs.

The approach of road damage detection based on deep learning proposed by Feng et al. [4] involved two-level processing. This hybrid method created is a two-step approach where YOLOv4 is used for the first detection of cracks, then DeepLabv3+ for segmentation improving the accuracy of cracks detection, especially when there is a lot of background noise that would obscure roads' damages. This approach increased the computational complexity.

Sami et al. [5] enhanced YOLOv5 for real-time road damage detection as follows: **Efficient Channel Attention (ECA-Net)** and **Focal Loss**. These modifications rectified the problem of class imbalance and helped the model to find concealed areas of the damage, for instance small fractures.

Transformer-based architectures received much attention because of modeling both local and global features for images. Chen et al. [6] proposed an **automatic pavement crack detection transformer** based on CNN and transformers. This ASL network combined the sequential and convolutional elements improving the ability to recognize intricate surface cracks while minimizing the computational load.

Feng et al. [7] worked on the problem of road crack detection using **Residual Convolutional Network**. Residual networks proved critical on deeper forms of the architecture since it made it possible for the network to learn more enhanced representations than shrouded in vanishing gradient problems. The enhancement shown by the addition of residual connections helped the model to detect cracks.

Hacıefendioglu et al. [8] utilized the **Faster R-CNN** for the detection of road crack, where RPN was applied to produce possible regions of interest. This improved localization of cracks and potholes compared favorably with other single-stage detectors using the two-stage detection approach.

Zhang et al. [9] proposed an enhanced version of **YOLOv3**, which is specially designed to detect small cracks. Enhancing the anchor boxes and increasing the input density, localization precision enhanced while recognizing fine details missed by prior YOLO versions.

He et al. [10] developed an **attention based road crack detection model using transformers**. This model utilized attention maps, for highlighting the important areas of the image virtually eliminating unnecessary background noise. The improved accuracy, found by the model, is attributed to enhanced guidance on regions which may harbor crack surfaces.

In the subsequent study, Lin et al. [11] proposed a **road crack detection transformer architecture** that strengthened feature extracting capability through fostered hierarchal structures. This architecture enabled the improved detection of features at both local and global scales and was more effective in difficult circumstances.

Chen et al. [12] proposed a **transformer encoder-decoder-based framework** for the crack detection that enables the model to address both the higher order and lower order texture features. The encoder-decoder have enhanced the general performance of the model because it was able to identify tiny and intricate cracks on road surfaces.

To the best of our knowledge, Arya et al. [13] presented **RDD2022**, a multi-national image dataset for road damage identification. This dataset included road images from various countries where each country offered different road damage and environment types. In this context, the fittings of RDD2022 have been helpful in assessing the transportability of the road damage detection models.

In response to the problem of class imbalance, Wan et al. [14] introduced **simplified YOLOv5 called YOLO-LRDD** as well as the **Focal Loss**, which would increase the quantity of identifiable rare kinds of damage. Focal Loss assisted in dealing with less frequent patterns of damage, for example minor cracking, which normal best practice losses hide from due to their occurrence in comparison to simple pothole damages.

Zeng et al. [15] proposed **YOLOv8-PD**, as a real-time road damage detection enhacement over the YOLOv8n. YOLOv8-PD used attention mechanisms and slim modules to reduce computational efforts while at the same time increasing detection precision. This model showed that architectures based on YOLO enable achieving high speed.

Apart from the conventional identification of road damage, Vokhidov et al. [16] also proposed a **CNN-based model** to identify damaged arrow-road markings using visible light camera sensor. This approach was useful for applications in IMS real time personal mobility and transportation systems. For example, Xu et al. [17] designed a vision-based system for **pavement marking detection and condition assessment** and applied image processing algorithms to guarantee appropriate pavement maintenance.

Wei et al. [18] proposed a method for **inspection of damages of road markings**, in which hierarchical semantic segmentation was employed along with dynamic homography estimation for accurate identification of damages with respect

TABLE I: Comparison of Studies Using YOLO, CNN, and EfficientNet

Study	Model/Method	Key Contributions	Limitations
Diao et al. [2]	LE-YOLOv5 (Lightweight YOLOv5)	Integrated attention mechanisms, optimized for mobile/edge devices, improved accuracy for small cracks.	Difficulty detecting fine details in complex environments.
Feng et al. [4]	YOLOv4 + DeepLabv3+	Hybrid model for initial detection and segmentation, effective in noisy environments.	Increased computational complexity.
Sami et al. [5]	YOLOv5 + ECA-Net + Focal Loss	Addresses class imbalance, enhances feature extraction for small objects, improves mAP.	None specified.
Zhang et al. [9]	Enhanced YOLOv3	Optimized for small cracks, better performance in complex backgrounds.	Challenges with lighting and occlusions.
Wan et al. [14]	YOLO-LRDD (Lightweight YOLOv5) + Focal Loss	Focuses on rare damage types, lightweight model for real-time applications.	None specified.
Zeng et al. [15]	YOLOv8-PD (Improved YOLOv8n)	Incorporates attention mechanisms, lightweight modules, ideal for real-time road damage detection.	None specified.
Arya et al. [19]	Deep Learning for Multinational Road Damage (YOLO)	Generalization across different environments, demonstrating YOLO's adaptability for global road management.	None specified.
Vokhidov et al. [17]	CNN-based Road Marking Detection	Effective for real-time detection of damaged road markings in intelligent transportation systems.	None specified.
Arya et al. [14]	RDD2022 (YOLO)	Multi-national dataset for evaluating YOLO-based road damage models across various geographical regions.	Class imbalance remains a challenge.

to various environmental conditions. This method provided a reliable approache to road marking inspections in actual field applications.

Finally, Arya et al. [19] , used deep learning for **multinational road damage detection** to establish generalization for different road states. This work highlighted the crucial need to build models that can cope with different scenarios.

The literature presented in the work reflects the progress achieved when using various deep learning techniques for road damage detection.

Table I shows summarized comparison of key studies that have employed YOLO, CNN, and EfficientNet in the domain of road damage detection:

III. METHODOLOGY

In this part, the detailed methodology as shown in Figure 2 of applying three deep learning models, which are **EfficientNet**, **Convolutional Neural Network (CNN)**, and **YOLOv8** for road damage detection, is described. The training process was performed using the **Roboflow Road Damage Detection dataset**.

A. Dataset Description

The Road Damage Detection dataset, sourced from Roboflow Universe, includes 8,586 images divided into 5,994 training, 1,797 validation, and 795 test images. The dataset focuses on detecting following seven types of road damage:

- **D00**: Longitudinal cracks
- **D10**: Transverse cracks
- **D20**: Alligator cracks
- **D40**: Potholes
- **D43**: Pavement edge breaks
- **D44**: Rutting
- **D50**: Other road damage

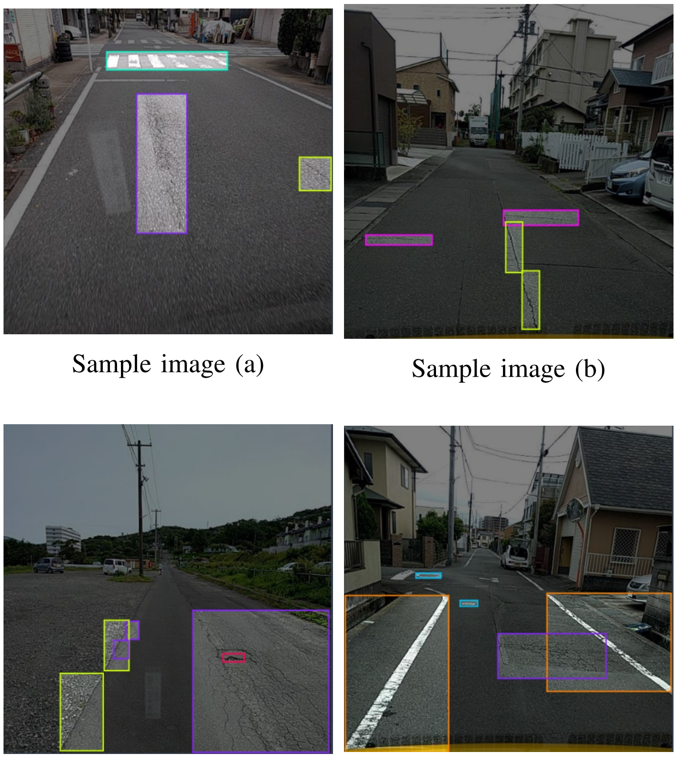
As shown in Figure 1, the dataset contains various images representing different types of road damage. All the images were preprocessed by resizing them to 640, 640 pixels to standardize the input for all models.The dataset contains challenges such as class imbalance, varying environmental conditions, and complex scenes with multiple overlapping damages type. Another challenge is the variation in images, which were captured under different weather conditions (sunny, cloudy, rainy) and lighting environments. This variability made it harder for the models to perform consistently. To mitigate these problems, during model training, we used data augmentation such as random flipping, rotation and brightness change.

B. EfficientNet Model

1) Model Architecture: EfficientNet Model is a family of convolutional neural networks created for high efficiency and scalability. In this study, **EfficientNet-B0** was used as it balances between performance and computational cost. This model achieves compound scaling to balance network depth, width, and resolution.

A fine tuned pre-trained EfficientNet-B0 model was used for this particular task by changing the final fully connected layer for classification of images into one of the seven types of road damages. This enabled the model not only to take advantage of the image features learned on ImageNet, yet also fine-tune on the task of road damage classification setup.

2) Training Process: The training was done using **American Adam Optimizer**, which is advantageous in working with sparse gradients and noisy data. The training was conducted using **20 epochs** and the batch size of the training process was **32**. The cross-entropy loss function was used, which is optimal when working with multi-class classification tasks. The learning rate set to $1e-4$, the learning rate tuning was planned to enhance the model performance.



Sample image (a) Sample image (b)

Sample image (c) Sample image (d)

Fig. 1: Sample Images from Road Damage Detection Dataset

C. Convolutional Neural Network (CNN)

1) Model Architecture: A new road damage detection **Convolutional Neural Network (CNN)** was developed. This model features **three convolutional layers (CONV)**, each paired with max-pooling layers to reduce the size of the feature maps. The convolutional layers progressively extract features from the input image, with the first layer focusing on basic details like edges and textures, while the deeper layers capture more complex patterns related to road damage.

After the convolutional layers, the feature maps are flattened and passed through a fully connected layer. To prevent overfitting, an **anti-overfitting layer** was added, which randomly drops some input units during each forward pass, helping to improve the model's generalization.

2) Training Process: The CNN model was trained using the Adam optimizer for **15 epochs**, with a learning rate of **0.001** and a batch size of **16**. Cross-entropy loss was applied for classification. During training, we encountered some images with missing or empty labels, which were addressed by implementing a custom collate function to filter out any invalid data.

D. YOLOv8 Model

1) Model Architecture: **YOLOv8** (You Only Look Once, Version 8) was utilized for both detecting and localizing objects within images. YOLOv8 is designed to identify multiple objects in a single image, drawing bounding boxes around them. It uses a unified neural network to handle both detection and classification tasks, leveraging feature maps at different scales to identify objects of various sizes.

2) Training Process: YOLOv8 was trained on the road damage dataset, using **predefined anchor boxes** to help the model predict bounding box coordinates and class labels. The model was trained for **15 epochs** with a batch size of **16**, and the process was run on a GPU to speed up the training. The learning process involved making predictions for both the bounding boxes and the respective class labels.

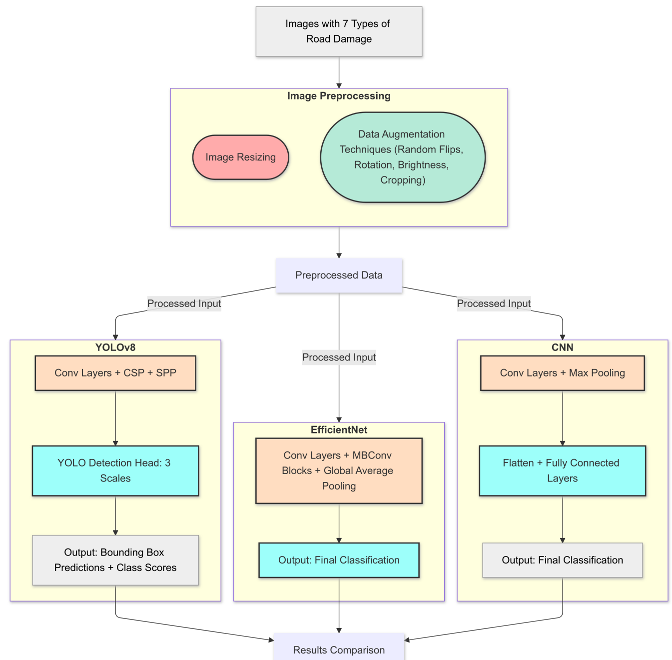


Fig. 2: Methodology

E. Summary of Experimental Setup

All images were resized to 640x640 pixels. Data augmentation techniques, including random flipping, rotation, and brightness adjustments, were applied. The CNN model was trained for 15 epochs with a batch size of 16, using the Adam optimizer and cross-entropy loss. EfficientNet was trained for 20 epochs with a batch size of 32 and a learning rate of 1e-4. YOLOv8, using predefined anchor boxes, was trained for 15 epochs with a batch size of 16..

IV. Evaluation, Results, and Discussion

A. Evaluation Metrics

To assess the performance of the models, we employed key evaluation metrics, including Accuracy, Precision, Recall, and F1-Score, providing a comprehensive view of each model's ability to classify road damage types. Additionally, we evaluated Mean Average Precision (mAP) to quantify the effectiveness of each model in detecting and correctly classifying multiple damage types across various confidence thresholds. **Precision** measures the proportion of correctly identified positive instances:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

Recall measures the proportion of actual positives that were correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

The **F1-Score** provides a balance between Precision and Recall:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Each model's performance was compared across multiple damage classes: **D00 (Longitudinal Cracks)**, **D10 (Transverse Cracks)**, **D20 (Alligator Cracks)**, **D40 (Potholes)**, **D43 (Pavement Edge Breaks)**, **D44 (Rutting)**, and **D50 (Other Damage)**. These categories served as benchmarks to evaluate the models' performance across different road damage types.

B. Results

1) Accuracy Comparison: The overall accuracy of each model is presented in Figure 3. **YOLOv8** achieved the highest accuracy with **64%**, followed by **EfficientNet** with **52%**, and **CNN** with **38%**. These values indicate the models' effectiveness in correctly identifying road damage categories.

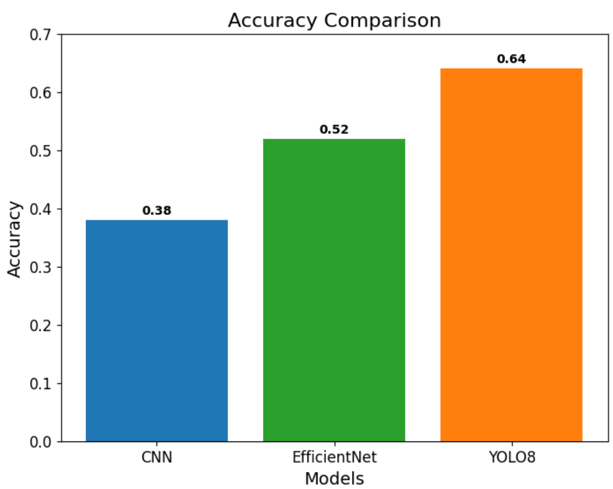


Fig. 3: Accuracy Comparison

2) F1-Score, Precision, and Recall Comparison: Figure 4 provides a detailed comparison of the **F1-Scores** across all damage categories. **YOLOv8** consistently outperformed the other models, especially in the classes **D00, D10, and D20**, where it achieved scores of **0.70, 0.69, and 0.68**, respectively. **EfficientNet** displayed moderate performance, particularly in simpler categories like **D00** and **D10**, while **CNN** lagged behind across most categories, especially **D43** and **D50**.

Figure 5 display the **Precision** and **Recall** comparisons. **YOLOv8** demonstrated superior performance in both metrics, particularly in damage types such as **D00, D43**, and **D44**, where high Precision and Recall scores were observed.

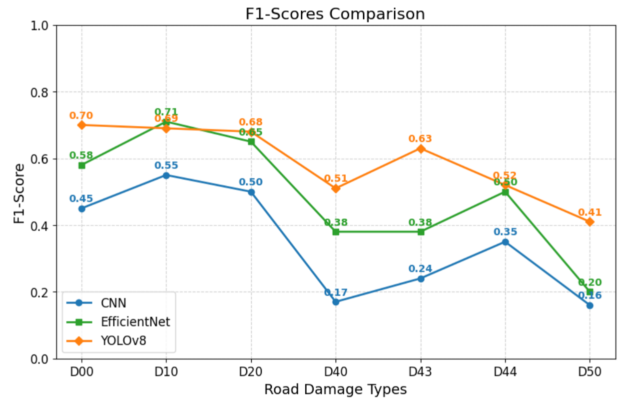


Fig. 4: F1 Score Comparison across Classes

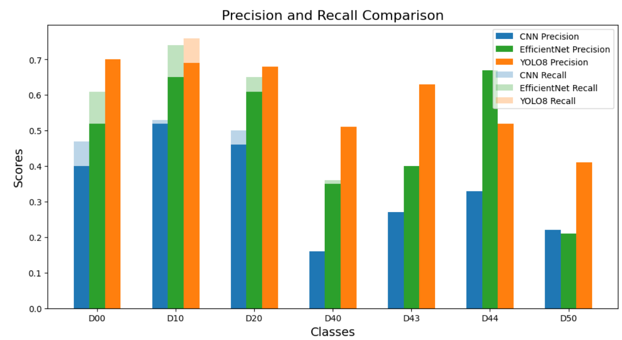


Fig. 5: Precision and Recall Comparison across Classes

3) Mean Average Precision (mAP): We also calculated the **Mean Average Precision (mAP)**, as shown in Figure 6, to evaluate the models' ability to detect road damage with varying levels of confidence. **YOLOv8** achieved the highest mAP score of **0.59**, significantly outperforming **EfficientNet (0.46)** and **CNN (0.34)**. These results further validate YOLOv8's robustness in both localization and classification of road damage.

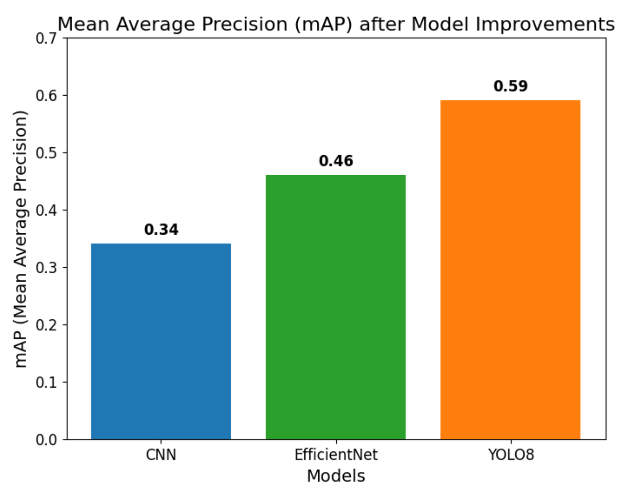


Fig. 6: Mean Average Precision (mAP) Comparison

Summarized results shown in Table II

TABLE II: Comparison of performance metrics for CNN, EfficientNet, and YOLOv8.

Model	Accuracy	Precision	Recall	F1-Score
EfficientNet	52%	0.52	0.52	0.51
CNN	38%	0.36	0.38	0.37
YOLOv8	64%	0.58	0.58	0.57

The radar plot in Figure 7 shows the performance comparison of the three models.

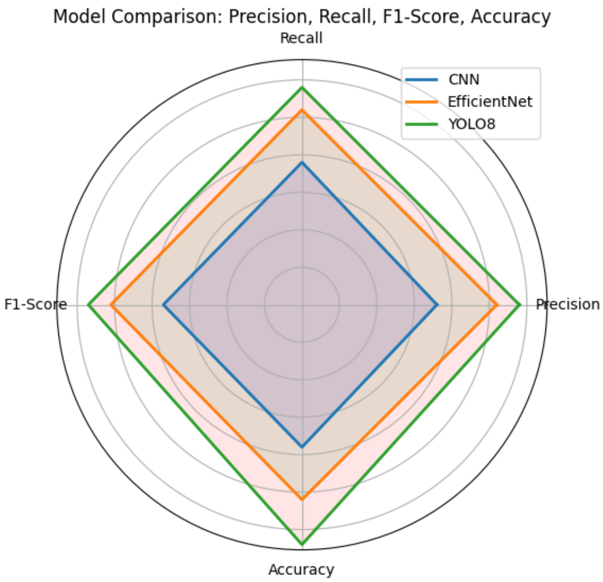


Fig. 7: Model Comparison: Precision, Recall, F1-Score, and Accuracy for CNN, EfficientNet, and YOLOv8.

C. Discussion

The results clearly demonstrate the superiority of **YOLOv8** over both **EfficientNet** and **CNN** in road damage detection. With an overall accuracy of **64%** and a mAP score of **0.59**, YOLOv8 is particularly adept at identifying complex damage types like **D00 (Longitudinal Cracks)** and **D20 (Alligator Cracks)**, which are critical for road maintenance and safety.

EfficientNet, with an accuracy of **52%** and mAP of **0.46**, showed moderate performance but struggled in detecting more complex damage types, such as **D43** and **D50**. **CNN**, with the lowest accuracy at **38%**, demonstrated limited capacity, performing best in simpler categories like **D10 (Transverse Cracks)** but faltering in more complex classes.

A common challenge faced by all models was distinguishing between visually similar classes such as **D43 (Pavement Edge Breaks)** and **D44 (Rutting)**, likely due to the low inter-class variance present in the dataset. This led to frequent classification errors.

V. Conclusion

In this study, we evaluated and compared three deep learning models—**CNN**, **EfficientNet**, and **YOLOv8**—for the task of road damage detection across multiple damage categories. The results demonstrate that **YOLOv8** outperforms the other models in terms of **accuracy**, **F1-Score**, **Precision**, **Recall**, and **Mean Average Precision (mAP)**. With a final accuracy of **64%** and a mAP of **0.59**, YOLOv8 proved to be the most reliable model for both classification and localization of road damage, especially in critical damage categories such as **D00 (Longitudinal Cracks)** and **D20 (Alligator Cracks)**.

Although **EfficientNet** achieved a moderate accuracy of **52%**, its mAP score of **0.46** indicates that it is less suitable for real-time road damage detection when compared to YOLOv8.

One of the major challenges across all models was the difficulty in distinguishing visually similar classes such as **D43 (Pavement Edge Breaks)** and **D44 (Rutting)**, likely due to the low inter-class variance in the dataset.

Future research should focus on further optimizing model architectures through **transfer learning** and enhanced **data augmentation techniques**.

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